CONSTRUCTION OF AN ABSOLUTELY INTEGRALLY CLOSED EXTENSION THAT IS FREE AS A MODULE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Absolutely integrally closed ring, meaning every monic polynomial decomposes into linear factors. defined at source lines 3509-3515 of the same chapter, before the counted statement. Supplies the property that the constructed ring must have.
- The functor adjoining a formal root of every monic polynomial, free on the given ring and a filtered colimit of finite free algebras. constructed at proof lines 3580-3590. Is the building block of the colimit.
- A monic polynomial over a filtered colimit of rings has its coefficients in some stage, and a root in the next stage survives to the colimit. used at proof lines 3595-3600. Is the argument that the colimit is absolutely integrally closed.

Source reference. Original lemma 0DCR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16741–16741. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0DCM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16737–16737. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *For any ring A there exists an extension $A \subset B$ such that*

(1) *B is a filtered colimit of finite free A-algebras,*

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(2) *B is free as an A-module, and*
(3) *B is absolutely integrally closed.*

Proof. Let I be the set of monic polynomials over A. For $i \in I$ denote x_i a variable and P_i the corresponding monic polynomial in the variable x_i. Then we set

$$F(A) = A[x_i; i \in I]/(P_i; i \in I)$$

As the notation suggests F is a functor from the category of rings to itself. Note that $A \subset F(A)$, that $F(A)$ is free as an A-module, and that $F(A)$ is a filtered colimit of finite free A-algebras. Then we take

$$B = \operatorname{colim} F^n(A)$$

where the transition maps are the inclusions $F^n(A) \subset F(F^n(A)) = F^{n+1}(A)$. Any monic polynomial with coefficients in B actually has coefficients in $F^n(A)$ for some n and hence has a solution in $F^{n+1}(A)$ by construction. This implies that B is absolutely integrally closed by Lemma 0DCM. We omit the proof of the other properties. □